Lab Eight – Probability

1 The Binomial and Negative Binomial Distributions

1.1 The Binomial Distribution

In many introductory courses, we make the simplifying assumption of independence in Bernoulli trials. For example, when playing best two-out-of-three legs in a darts match, we would assume Player A has an 70% chance of winning each leg. We write:

$$X \sim \text{Binomial}(n = 3, p = 0.70)$$

where

- X is the number of legs won
- n is the number of trials
- p is the probability of the event of interest

Of note, it is not necessarily the case that the players play all three legs – if either player wins legs one and two the match ends.

There are eight possible outcomes for Player A:

- (win, win, "win") $\longrightarrow$ Player A wins
- (win, win, "lose") $\longrightarrow$ Player A wins
- (win, lose, win) $\longrightarrow$ Player A wins
- (win, lose, lose) $\longrightarrow$ Player A loses
- (lose, win, win) $\longrightarrow$ Player A wins
- (lose, win, lose) $\longrightarrow$ Player A loses
- (lose, lose, "win") $\longrightarrow$ Player A loses
- (lose, lose, "lose") $\longrightarrow$ Player A loses

Note: I use quotations above to denote outcomes where the last match isn't actually played.

Use the Binomial distribution to compute the probability that Player A wins the match. Make sure write out your probability statement and work to write it in terms of the distribution function (PMF/CDF) to plot the probability using `ggplot`.

Solution

```r
install.packages("tidyverse")
```

```r
library("tidyverse")
```

The probability that Player A wins the match is the probability of winning two or more legs.

$P(X \geq 2) = f_X(2) + f_X(3) = 0.784$

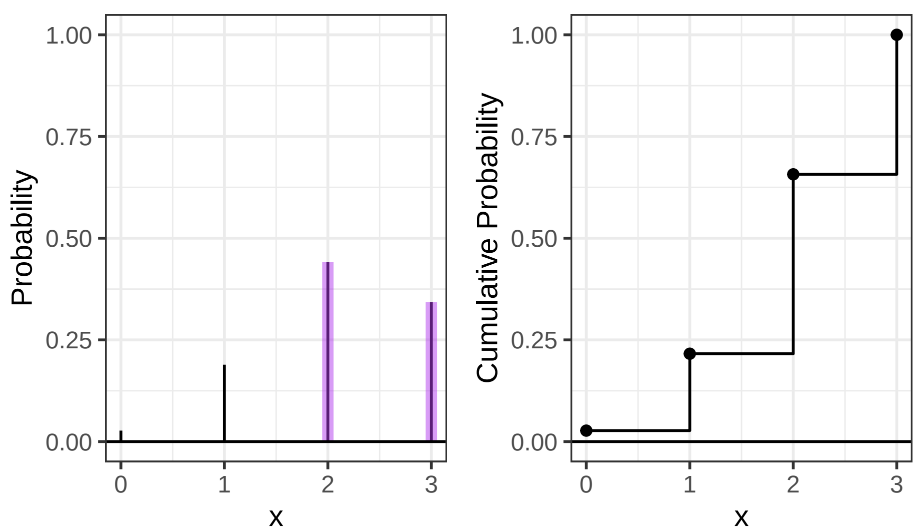
```r
(dbinom(x=2, prob = 0.70, size = 3) + dbinom(x=3, prob = 0.70, size = 3))
```

```
[1] 0.784
```

```r
ggdat <- tibble(x = 0:3) |>
  mutate(PMF = dbinom(x, prob = 0.70, size = 3),
         CDF = pbinom(x, prob = 0.70, size = 3))

PMF.plot <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
```

```
  geom_linerange(data = ggdat |> filter(x >= 2),
                 aes(x = x, ymin = 0, ymax = PMF),
                 color = "darkorchid2", linewidth = 2, alpha = 0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0, 1) +
  ylab("Probability") +
  theme_bw()

CDF.plot <- ggplot() +
  geom_step(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Cumulative Probability") +
  theme_bw()

PMF.plot | CDF.plot
```

Figure 1: The probability that Player A wins the match



1.2 The Negative Binomial Distribution

In intermediate courses, we learn the Negative Binomial distribution that we use to model the number of "failures" until the rth success. Note that this approach still makes the simplifying assumption of independence in Bernoulli trials.

Specify the two cases necessary to compute the probability. Fully explain these probability statements in the context of a best two-out-of-three match and the Negative Binomial distribution. Finally, compute the probability that Player A wins the match and show you get the same result as the Binomial approach. Make sure write out your probability statement and work to write it in terms of the distribution function (PMF/CDF) to plot the probability using `ggplot`.

Solution

The two cases where Player A wins is when they have 1 loss or no loss before their 2nd success: (lose, win, win) or (win, lose, win).

$P(X < 2) = f_X(1) + f_X(0) = F_X(1) = 0.784$

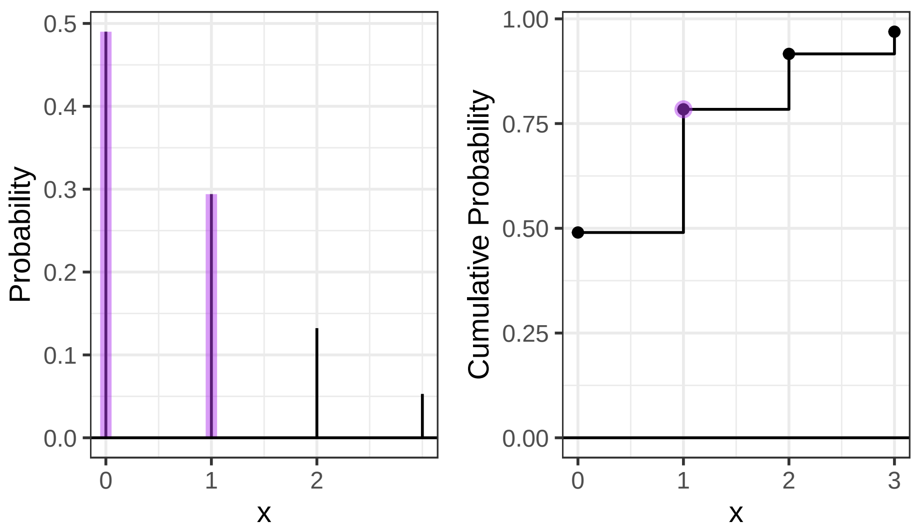
```
(dnbinom(x=1, prob = 0.70, size = 2)+ dnbinom(x=0, prob = 0.70, size = 2))
```

```
[1] 0.784
```

```
ggdat <- tibble(x = 0:3) |>
  mutate(PMF = dnbinom(x, prob = 0.70, size = 2),
         CDF = pnbinom(x, prob = 0.70, size = 2))

```

```r
PMF.plot <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_linerange(data = ggdat |> filter(x < 2),
                 aes(x = x, ymin = 0, ymax = PMF),
                 color = "darkorchid2", linewidth = 2, alpha = 0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:2) +
  ylab("Probability") +
  theme_bw()

CDF.plot <- ggplot() +
  geom_step(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat |> filter(x==1),
             aes(x = x, y = CDF),
             color="darkorchid2", size=2.5, alpha=0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Cumulative Probability") +
  theme_bw()

PMF.plot | CDF.plot
```

Figure 2: The probability that Player A wins the match



2 The Beta Binomial Distribution

One approach to generalize this modeling is to use a Beta-Binomial distribution. This distribution arrises when we let p be a random variable that follows a Beta distribution:

$$P \sim \text{Beta}(\alpha, \beta)$$
$$X \sim \text{Binomial}(n, P).$$

This distribution is not part of base `R`, but is available using the `_bbinom` functions from the `extraDistr` package (Wolodzko 2026).

The primary benefit is that the Beta-Binomial enables us to specify some uncertainty about the skill of Player A, as p does not have to be fixed (e.g., $p = 0.70$). Suppose the result of the best two-out-of-three match follows a Beta-Binomial distribution where $\alpha = 7$ and $\beta = 3$.

2.1 The Beta Part

1. **Plot the corresponding Beta distribution. Interpret the plot in the context of a best two-out-of-three match and the Binomial distribution.**
   Hint: The Beta distribution is available in base `R` via `_beta()`.

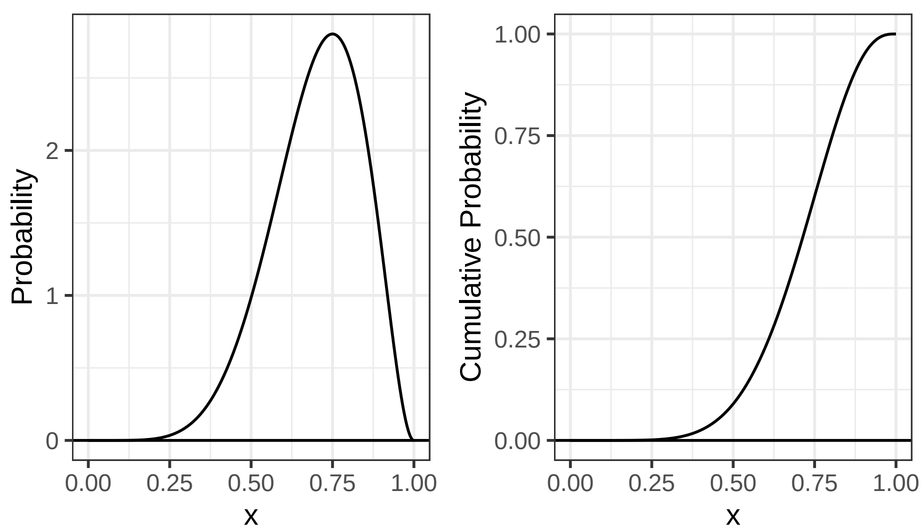
Solution

```
install.packages("extraDistr")
```

```
library("extraDistr")
```

```
ggdat <- tibble(x = seq(0, 1, length.out=1000))  |>
  mutate(PDF = dbeta(x, 7, 3,
                      log = FALSE),
        CDF = pbeta(x, 7, 3, lower.tail = TRUE,
                      log.p = FALSE))

PDF.plot <- ggplot() +
  geom_line(data = ggdat,
               aes(x = x, y = PDF)) +
  geom_hline(yintercept = 0) +
  ylab("Probability") +
  theme_bw()

CDF.plot <- ggplot() +
  geom_line(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
  ylab("Cumulative Probability") +
  theme_bw()

PDF.plot | CDF.plot
```

Figure 3: The probability that Player A wins the match



The probability of Player A winning a leg cluster around 0.70, meaning on average, Player A still wins 70% of the time. The distribution is left-skewed, i.e.,there are more higher values than lower values, meaning the probability of Player A winning 1 leg is generally higher around $0.70 - 0.80$.

2. How does the Beta incorporate uncertainty about the skill of Player A? **Compute the mean and variance of the distribution.**
   Hint: You will find the `integrate(...)` function helpful here.

$$E(P) = \mu_P = \int_0^1 p f_P(p) dp$$

$$var(P) = \sigma_P^2 = \int_0^1 (p - \mu_P)^2 f_P(p) dp$$

Solution The Beta incorporate uncertainty about the skill of Player A because it treats the probability of winning as a variable with a distribution (rather than a constant).

$E(P) = 0.7, var(P) = 0.019$

```
integrand_mean = function(p) {
  p*(dbeta(p, 7, 3, log = FALSE))
  }
(mean=integrate(integrand_mean, 0, 1))
```

```
0.7 with absolute error < 7.8e-15
```

```
integrand_variance = function(p) {
  ((p-mean$value)^2)*(dbeta(p, 7, 3, log = FALSE))
  }
(variance=integrate(integrand_variance, 0, 1))
```

```
0.01909091 with absolute error < 2.1e-16
```

3. **What is the probability that Player A has a win probability of exactly 0.70? Make sure write out your probability statement and work to write it in terms of the distribution function (PDF/CDF) to plot the probability using `ggplot`.**
   Hint: Think carefully about the properties of continuous distributions here.

Solution

The probability of Player A having a win probability of exactly 0.70 is 0. Under a continuous distribution, a unit can always be made finer (i.e., a variable can take on an infinite number of possible values within any interval).

4. **What is the probability that Player A has a win probability of at least 0.50? Make sure write out your probability statement and work to write it in terms of the distribution function (PDF/CDF) to plot the probability using `ggplot`.**

Solution

The probability of Player A having a win probability of at least 0.5 is

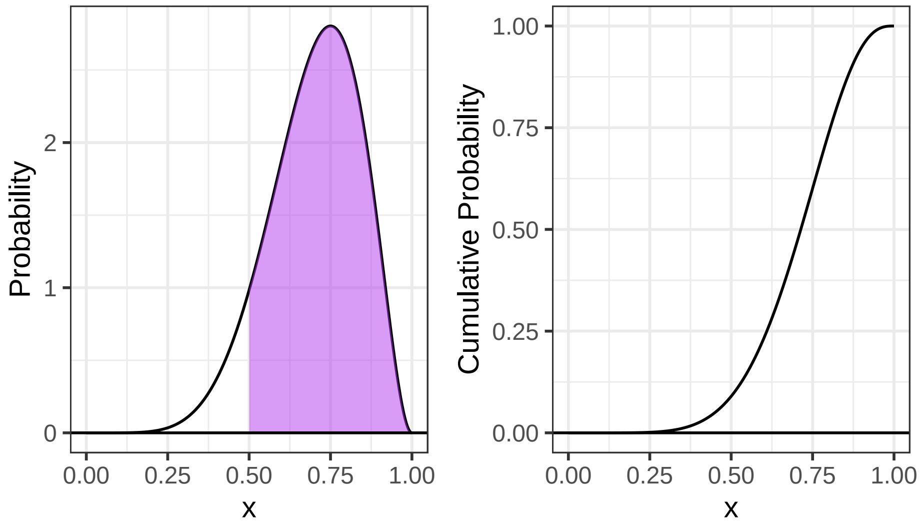
$$P(p \geq 0.5) = P(p > 0.5) = 1 - F_x(0.5) = 0.91$$

```
(1-pbeta(0.5, 7, 3))
```

```
[1] 0.9101562
```

```
ggdat <- tibble(x = seq(0, 1, length.out=1000))|>
  mutate(PDF = dbeta(x, 7, 3,
                     log = FALSE),
        CDF = pbeta(x, 7, 3, lower.tail = TRUE,
                     log.p = FALSE))

PDF.plot <- ggplot() +
  geom_line(data = ggdat,
               aes(x = x, y = PDF)) +
  geom_ribbon(data = ggdat|> filter(x >=0.5),
              aes(x = x, ymin=0, ymax=PDF),
              fill="darkorchid2", alpha=0.50) +
  geom_hline(yintercept = 0) +
  ylab("Probability") +
  theme_bw()

CDF.plot <- ggplot() +
  geom_line(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
  ylab("Cumulative Probability") +
  theme_bw()

PDF.plot | CDF.plot
```

Figure 4: The probability that Player A has a win probability of at least 0.5



5. **What is the 90th percentile of Player A's win probability based on this Beta distribution? Make sure write out your probability statement and work to write it in terms of the distribution function (PDF/CDF) to plot the percentile using `ggplot`.**

Solution

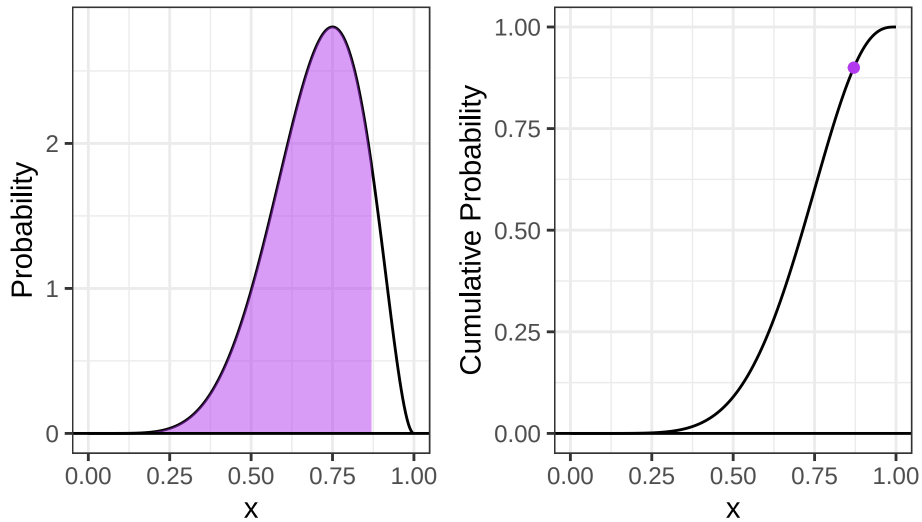
The 90th percentile of Player A's win probability is $F_X^{-1}(0.9) = 0.87$.

```
(qbeta(0.9, 7, 3))
```

```
[1] 0.8705027
```

```
ggdat <- tibble(x = seq(0, 1, length.out=1000))|>
  mutate(PDF = dbeta(x, 7, 3,
                     log = FALSE),
         CDF = pbeta(x, 7, 3, lower.tail = TRUE,
                     log.p = FALSE))

PDF.plot <- ggplot() +
  geom_line(data = ggdat, aes(x = x, y = PDF)) +
  geom_ribbon(data = ggdat |> filter(x <= qbeta(0.9, 7, 3)),
              aes(x = x, ymin = 0, ymax = PDF),
              fill = "darkorchid2", alpha = 0.5) +
  geom_hline(yintercept = 0) +
  ylab("Probability") +
  theme_bw()

CDF.plot <- ggplot() +
  geom_line(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = tibble(x = qbeta(0.9, 7, 3))|> mutate(CDF = pbeta(x, 7, 3)),
             aes(x = x, y=CDF), color="darkorchid2")+
  geom_hline(yintercept = 0) +
  ylab("Cumulative Probability") +
  theme_bw()

PDF.plot | CDF.plot
```

Figure 5: The 90th percentile of Player A's win probability



2.2 The Beta-Binomial Part

Compute the probability that Player A wins the match. Compare your result to the Binomial and Negative Binomial results. Make sure write out your probability statement and work to write it in terms of the distribution function (PMF/CDF) to plot the probability using `ggplot`

Solution

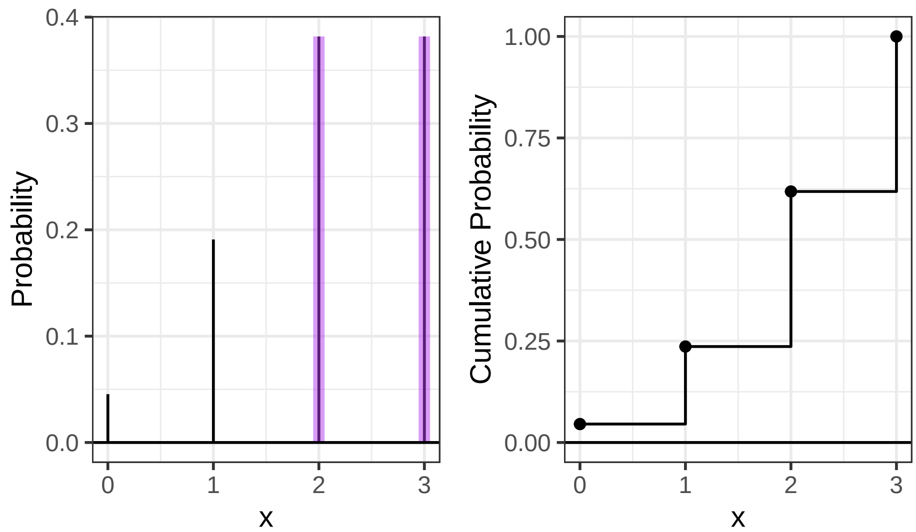
The probability of Player A winning the match is $P(X \geq 2) = 0.76$. This is lower than the results from the binomial and negative binomial distribution, because of the uncertainty about the skill of Player A.

```r
(dbbinom(2,3,7,3) + dbbinom(3,3,7,3))
```

```
[1] 0.7636364
```

```r
ggdat <- tibble(x = 0:3) |>
  mutate(PMF = dbbinom(x,3,7,3),
         CDF=pbbinom(x,3,7,3))

PMF.plot <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_linerange(data = ggdat |> filter(x >= 2),
                 aes(x = x, ymin = 0, ymax = PMF),
                 color = "darkorchid2", linewidth = 2, alpha = 0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Probability") +
  theme_bw()

CDF.plot <- ggplot() +
  geom_step(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Cumulative Probability") +
  theme_bw()

PMF.plot | CDF.plot
```

Figure 6: The probability that Player A wins the match, considering uncertainty in skills



2.3 The Why

Check the expected value and variance of each distribution.

Analytically, we have

$$E(X) = np \qquad var(X) = np(1-p) \qquad \text{(Binomial Distribution)}$$

$$E(X) = n\left(\frac{\alpha}{\alpha+\beta}\right) \qquad var(X) = \frac{n\alpha\beta(\alpha+\beta+n)}{(\alpha+\beta)^2(\alpha+\beta+1)} \qquad \text{(Beta-Binomial Distribution)}$$

For each distribution, compute the mean number of legs won for player A.
Hint: You can perform the sum over the finite support set here.

$$E(X) = \mu_X = \sum_{i=0}^{3} i f_X(i)$$
$$var(X) = \sigma_X = \sum_{i=0}^{3} (i-\mu_X) f_X(i)$$

Solution

The two distributions have the same mean, but the Beta-Binomial Distribution has higher variance, which makes sense because it accounts for uncertainty in the skill of Player A.

- Binomial distribution: $E(X) = 3 \times 0.70 = 2.1, var(X) = 3 \times 0.70(1-0.70) = 0.63$
- Beta-Binomial Distribution: $E(X) = 0.70(\frac{7}{7+3}) = 2.1, var(X) = \frac{3\times3\times(3+7+3)}{(3+7)^2(3+7+1)} = 0.74$

Compute the mean of number of legs won for Player A.

- Binomial distribution:

```
# sum
x=0:3
(mean = sum(x*dbinom(x, prob = 0.70, size = 3)))
```

```
[1] 2.1
```

```
(var=sum((x-mean)^2*dbinom(x, prob = 0.70, size = 3)))
```

```
[1] 0.63
```

- Beta-binomial distribution

```
x=0:3
(mean = sum(x*dbbinom(x,3,7,3)))
```

```
[1] 2.1
```

```
(var=sum((x-mean)^2*dbbinom(x,3,7,3)))
```

```
[1] 0.7445455
```

3 Altham's Multiplicative Binomial Distribution

While the Beta-Binomial enables us to model cases where winning has a clustering effect (e.g., a "hot hand").

3.1 Probability Mass Function

The mass function for this distribution is below. While it used to be available in the `MBD` package for `R`, this package is no longer available.

$$f_X(x) = \frac{1}{c}\binom{n}{x}p^x(1-p)^{n-x}\theta^{x(n-x)}I(x \in \{0,1,...,n\})$$

where

$$c = \sum_{i=0}^{n}\binom{n}{i}p^i(1-p)^{n-i}\theta^{i(n-i)}$$

Write a function in `R` called `daltbinom()` for computing the PMF of this distribution. Use your function to plot the distribution where $p = 0.70$ where $\theta = 0.50$, $\theta = 1$, and $\theta = 1.50$. Comment on the differences in the context of a best two-out-of-three match

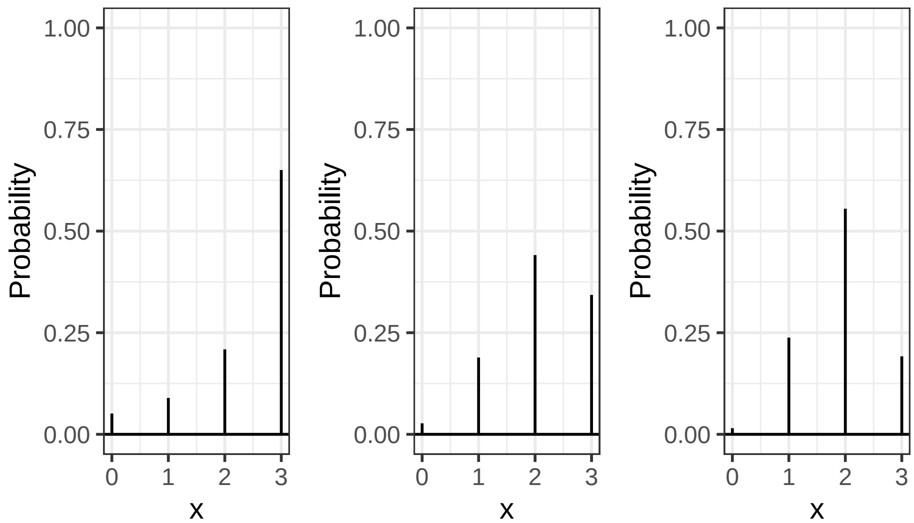
Solution

```
daltbinom = function(x, n, p, theta) {
  i = 0:n
  c = sum(choose(n, i)*p^i*(1-p)^(n-i)*theta^(i*(n-i)))
  (1/c)*choose(n, x)*p^x*(1-p)^(n-x)*theta^(x*(n-x))
}
```

```
ggdat <- tibble(x = 0:3) |>
  mutate(PMF = daltbinom(x,3,0.70,0.5))

PMF.plot.1 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0,1)+
  ylab("Probability") +
  theme_bw()

ggdat <- tibble(x = 0:3) |>
  mutate(PMF = daltbinom(x,3,0.70,1))

PMF.plot.2 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0,1)+
  ylab("Probability") +
  theme_bw()

ggdat <- tibble(x = 0:3) |>
  mutate(PMF = daltbinom(x,3,0.70,1.5))

PMF.plot.3 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0,1)+
```

```r
  ylab("Probability") +
  theme_bw()

PMF.plot.1 | PMF.plot.2 | PMF.plot.3
```

Figure 7: PMF of Altham's Multiplicative Binomial Distribution when θ=0.5, θ=1 and θ=1.5



When $\theta = 1$, it's a binomial distribution, which is a probability distribution of the number of leg Player A can win out of 3 trials that are independent. When θ is smaller than 1, the probability of Player A winning more legs after winning 1 leg is higher, so the probability of winning (WW"L" and WW"W") is significantly higher. When θ is larger than 1, Player A is more likely to win 2 trials out of 3 (2-1 wins are more likely than sweeps).

3.2 Cumulative Distribution Function

Use your `daltbinom()` function to create a `paltbinom()` function for computing the CDF of this distribution. Use your function to plot the distribution where $p = 0.70$ where $\theta = 0.50$, $\theta = 1$, and $\theta = 1.50$.

Solution

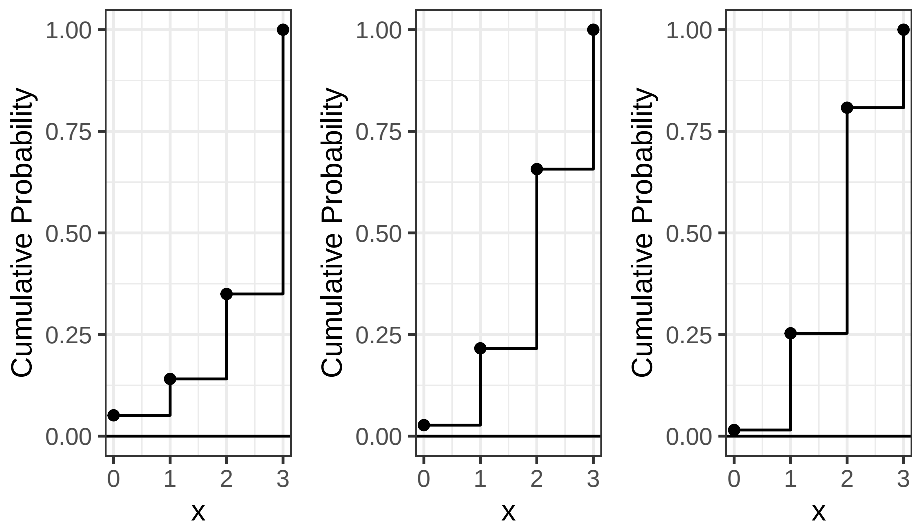
The CDF at point x is just the sum of all the PMF values from 0 through x.

```r
# vectorized
paltbinom = function(x, n, p, theta) {
  sum(daltbinom(0:x, n, p, theta))
}
```

```r
ggdat <- tibble(x = 0:3) |>
  rowwise() |>
  mutate(CDF = paltbinom(x,3,0.70,0.5))

CDF.plot.1 <- ggplot() +
  geom_step(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Cumulative Probability") +
  theme_bw()

ggdat <- tibble(x = 0:3) |>
  rowwise() |>
  mutate(CDF = paltbinom(x,3,0.70,1))

CDF.plot.2 <- ggplot() +
  geom_step(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
```

```
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Cumulative Probability") +
  theme_bw()

ggdat <- tibble(x = 0:3) |>
  rowwise() |>
  mutate(CDF = paltbinom(x,3,0.70,1.5))

CDF.plot.3 <- ggplot() +
  geom_step(data = ggdat, aes(x = x, y = CDF)) +
  geom_point(data = ggdat, aes(x = x, y = CDF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylab("Cumulative Probability") +
  theme_bw()

CDF.plot.1 | CDF.plot.2 | CDF.plot.3
```

Figure 8: CDF of Altham's Multiplicative Binomial Distribution when θ=0.5, θ=1 and θ=1.5



3.3 The Winning Probability

Compute the probability that Player A wins the match using Altham's Multiplicative Binomial Distribution. Compare your result to the Binomial and Negative Binomial results. Compare your results to the Beta-Binomial model, too. Make sure write out your probability statement and work to write it in terms of the distribution function (PMF/CDF) to plot the probability using `ggplot`.

Solution

The probability that Player A wins the match is the probability of winning two or more legs.

$$P(X \geq 2) = 1 - F_X(1)$$

For each distribution with different θ, this probability is:

```
(1-paltbinom (1,3,0.70,0.5))
```

```
[1] 0.8592417
```

```
(1-paltbinom (1,3,0.70,1))
```

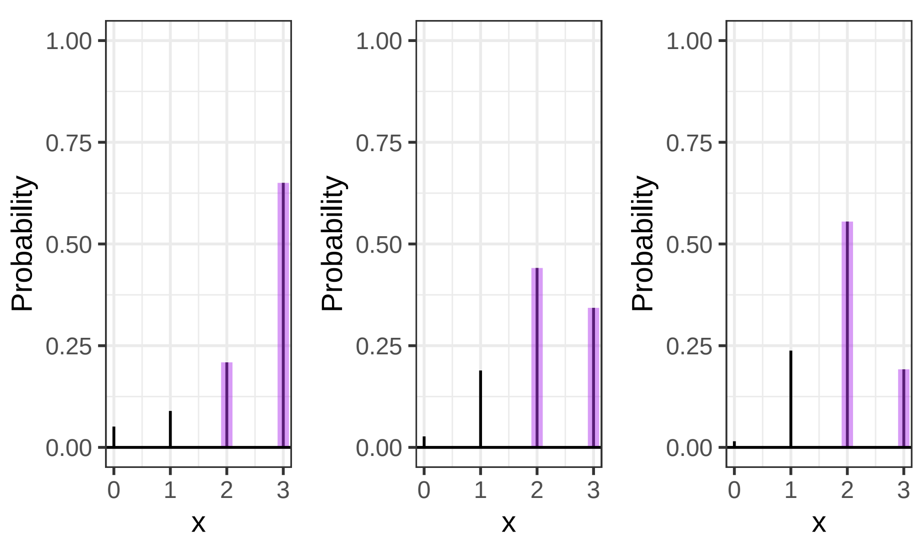
```
[1] 0.784
```

```
(1-paltbinom (1,3,0.70,1.5))
```

```
[1] 0.746993
```

```r
ggdat <- tibble(x = 0:3) |>
  mutate(PMF = daltbinom(x,3,0.70,0.5))

PMF.plot.1 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_linerange(data = ggdat |> filter(x >= 2),
                 aes(x = x, ymin = 0, ymax = PMF),
                 color = "darkorchid2", linewidth = 2, alpha = 0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0,1)+
  ylab("Probability") +
  theme_bw()

ggdat <- tibble(x = 0:3) |>
  mutate(PMF = daltbinom(x,3,0.70,1))

PMF.plot.2 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_linerange(data = ggdat |> filter(x >= 2),
                 aes(x = x, ymin = 0, ymax = PMF),
                 color = "darkorchid2", linewidth = 2, alpha = 0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0,1)+
  ylab("Probability") +
  theme_bw()

ggdat <- tibble(x = 0:3) |>
  mutate(PMF = daltbinom(x,3,0.70,1.5))

PMF.plot.3 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_linerange(data = ggdat |> filter(x >= 2),
                 aes(x = x, ymin = 0, ymax = PMF),
                 color = "darkorchid2", linewidth = 2, alpha = 0.50) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0,1)+
  ylab("Probability") +
  theme_bw()

PMF.plot.1 | PMF.plot.2 | PMF.plot.3
```

Figure 9: The probability that Player A wins the match when θ=0.5, θ=1 and θ=1.5



4 Altham's Multiplicative Beta-Binomial Distribution

You may have wondered – "Can we do the Beta thing with Altham's version of the Binomial Distribution?" The answer is yes!

4.1 Probability Mass Function

The mass function for this distribution is below.

$$f_X(x) = \left[\int_0^1 dp \left(\frac{1}{c}\binom{n}{x}p^x(1-p)^{n-x}\theta^{x(n-x)}\right)\left(\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}\right)\right] I(x \in \{0,1,...,n\})$$

Write a function in R called `daltbbinom()` for computing the PMF of this distribution. Use your function to plot the distribution where $\alpha = 7$ and $\beta = 3$ where $\theta = 0.50$, $\theta = 1$, and $\theta = 1.50$. Comment on the differences in the context of a best two-out-of-three match.

Hint: You can use the `integrate()` in R to compute the integral. Note that this requires the function to be vectorized – it should be unless you have added any block conditionals.

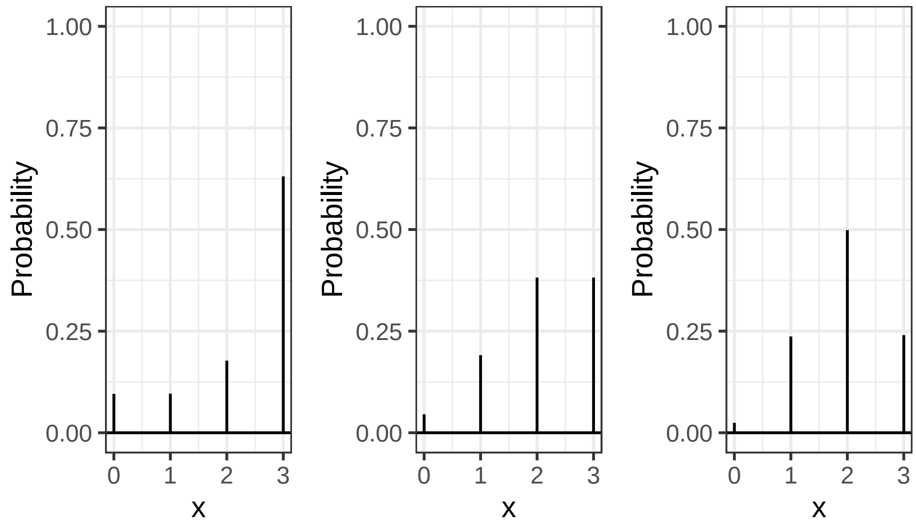
Solution

```
daltbbinom = function(x, n, alpha, beta, theta) {
  # have to vectorize to use integrate
  integrand = Vectorize(function(p){
    i=0:n
    c = sum(choose(n,i)*p^i*(1-p)^(n-i)*theta^(i*(n-i)))
    (1/c)*choose(n,x)*p^x*(1-p)^(n-x)*theta^(x*(n-x))*
      (gamma(alpha+beta)/(gamma(alpha)*gamma(beta)))*
      p^(alpha-1)*(1-p)^(beta-1)
  })
  integrate(integrand,0,1)$value
}
```

```
ggdat <- tibble(x=0:3)|>
  rowwise()|>
  mutate(PMF = daltbbinom(x,3,7,3,0.5))

PMF.plot.1 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0, 1) +
  ylab("Probability") +
  theme_bw()

ggdat <- tibble(x=0:3)|>
  rowwise()|>
  mutate(PMF = daltbbinom(x,3,7,3,1))

PMF.plot.2 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0, 1) +
  ylab("Probability") +
  theme_bw()

ggdat <- tibble(x=0:3)|>
  rowwise()|>
  mutate(PMF = daltbbinom(x,3,7,3,1.5))

PMF.plot.3 <- ggplot() +
  geom_linerange(data = ggdat,
                 aes(x = x, ymin = 0, ymax = PMF)) +
  geom_hline(yintercept = 0) +
  scale_x_continuous("x", breaks = 0:3) +
  ylim(0, 1) +
  ylab("Probability") +
  theme_bw()
```

```
PMF.plot.1 | PMF.plot.2 | PMF.plot.3
```

Figure 10: PMF of Altham's Multiplicative Beta-Binomial Distribution when θ=0.5, θ=1 and θ=1.5



When $\theta = 1$, it's a binomial distribution, assuming each leg is independent. When θ is smaller than 1, the probability of Player A winning the next leg after winning 1 leg is higher, so the probability of winning (WW"L" and WW"W") is significantly higher. When θ is larger than 1, Player A is more likely to win 2 trials out of 3 (2-1 wins are more likely).

5 A Comparison

Compute the probabilities for all outcomes for Player A.

- Loses 0-2 (LL"L", LL"W")
- Loses 1-2 (WLL, LWL)
- Wins 2-0 (WW"W", WW"L")
- Wins 2-1 (LWW, WLW)

Note: You'll need to define these outcomes in terms of the random variable.

Create a bar plot (or column plot) using `ggplot` to show the probability of each outcome under the following conditions:

- **Binomial**
- **Negative Binomial**
  Note: You can compute the lose probabilities the same way as the win probabilities using success probability $(1-p)$
- **Beta-Binomial**
- **Althman Binomial with** $\theta = 0.50$ **and** $\theta = 1.5$
- **Althman Beta-Binomial with** $\theta = 0.50$ **and** $\theta = 1.5$

Comment on the differences in the context of a best two-out-of-three match.

Solution Defining outcomes in terms of random variable

- Binomial:
  - Loses 0-2: $P(X=0) + \frac{1}{3}P(X=1) = f_X(0) + \frac{1}{3}f_X(1)$
  - Loses 1-2: $\frac{2}{3}P(X=1) = \frac{2}{3}f_X(1)$
  - Wins 2-0: $P(X=3) + \frac{1}{3}P(X=2) = f_X(3) + \frac{1}{3}f_X(2)$

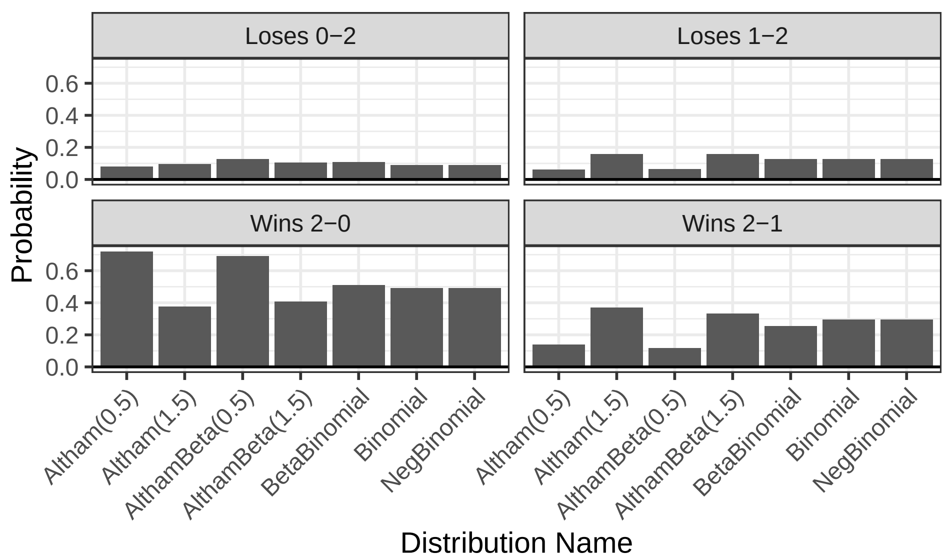
    - Wins 2-1: $\frac{2}{3}P(X=2) = \frac{2}{3}f_X(2)$
- Negative Binomial:
    - Wins 2-0: $P(X=0) = f_X(0)$
    - Wins 2-1: $P(X=1) = f_X(1)$
    - Loses: Using $p = 0.30$ instead of $p = 0.70$ (the probability of losing the leg), Loses 0-2 is $P(X=0)$ (0 successes before 2 failures) and Loses 1-2 is $P(X=1)$
- Beta-Binomial
    - Loses 0-2: $P(X=0) + \frac{1}{3}P(X=1) = f_X(0) + \frac{1}{3}f_X(1)$
    - Loses 1-2: $\frac{2}{3}P(X=1) = \frac{2}{3}f_X(1)$
    - Wins 2-0: $P(X=3) + \frac{1}{3}P(X=2) = f_X(3) + \frac{1}{3}f_X(2)$
    - Wins 2-1: $\frac{2}{3}P(X=2) = \frac{2}{3}f_X(2)$
- Althman Binomial:
    - Loses 0-2: $P(X=0) + \frac{1}{3}P(X=1) = f_X(0) + \frac{1}{3}f_X(1)$
    - Loses 1-2: $\frac{2}{3}P(X=1) = \frac{2}{3}f_X(1)$
    - Wins 2-0: $P(X=3) + \frac{1}{3}P(X=2) = f_X(3) + \frac{1}{3}f_X(2)$
    - Wins 2-1: $\frac{2}{3}P(X=2) = \frac{2}{3}f_X(2)$
- Althman Beta-Binomial
    - Loses 0-2: $P(X=0) + \frac{1}{3}P(X=1) = f_X(0) + \frac{1}{3}f_X(1)$
    - Loses 1-2: $\frac{2}{3}P(X=1) = \frac{2}{3}f_X(1)$
    - Wins 2-0: $P(X=3) + \frac{1}{3}P(X=2) = f_X(3) + \frac{1}{3}f_X(2)$
    - Wins 2-1: $\frac{2}{3}P(X=2) = \frac{2}{3}f_X(2)$

```
comparing <- tibble(
  outcome=rep(c("Loses 0-2", "Loses 1-2", "Wins 2-0", "Wins 2-1"),times=7),
  distribution=rep(c("Binomial",
                     "NegBinomial",
                     "BetaBinomial",
                     "Altham(0.5)",
                     "Altham(1.5)",
                     "AlthamBeta(0.5)",
                     "AlthamBeta(1.5)"),each=4),
  probability=c(
    # Binomial
    dbinom(0,3,0.70)+(1/3)*dbinom(1,3,0.70),
    (2/3)*dbinom(1,3,0.70),
    dbinom(3,3,0.70)+(1/3)*dbinom(2,3,0.70),
    (2/3)*dbinom(2,3,0.70),
    # Negative Binomial
    dnbinom(0,2,0.30), dnbinom(1,2,0.30), dnbinom(0,2,0.70), dnbinom(1,2,0.70),
    # Beta-Binomial
    dbbinom(0,3,7,3)+(1/3)*dbbinom(1,3,7,3),
    (2/3)*dbbinom(1,3,7,3),
    dbbinom(3,3,7,3)+(1/3)*dbbinom(2,3,7,3),
    (2/3)*dbbinom(2,3,7,3),
    # Altham Binomial theta=0.5
    daltbinom(0,3,0.70,0.5)+(1/3)*daltbinom(1,3,0.70,0.5),
    (2/3)*daltbinom(1,3,0.70,0.5),
    daltbinom(3,3,0.70,0.5)+(1/3)*daltbinom(2,3,0.70,0.5),
    (2/3)*daltbinom(2,3,0.70,0.5),
    # Altham Binomial theta=1.5
    daltbinom(0,3,0.70,1.5)+(1/3)*daltbinom(1,3,0.70,1.5),
    (2/3)*daltbinom(1,3,0.70,1.5),
    daltbinom(3,3,0.70,1.5)+(1/3)*daltbinom(2,3,0.70,1.5),
    (2/3)*daltbinom(2,3,0.70,1.5),
    # Altham Beta-Binomial theta=0.5
    daltbbinom(0,3,7,3,0.5)+(1/3)*daltbbinom(1,3,7,3,0.5),
```

```
    (2/3)*daltbbinom(1,3,7,3,0.5),
    daltbbinom(3,3,7,3,0.5)+(1/3)*daltbbinom(2,3,7,3,0.5),
    (2/3)*daltbbinom(2,3,7,3,0.5),
    # Altham Beta-Binomial theta=1.5
    daltbbinom(0,3,7,3,1.5)+(1/3)*daltbbinom(1,3,7,3,1.5),
    (2/3)*daltbbinom(1,3,7,3,1.5),
    daltbbinom(3,3,7,3,1.5)+(1/3)*daltbbinom(2,3,7,3,1.5),
    (2/3)*daltbbinom(2,3,7,3,1.5)
  )
)
```

```
ggplot(comparing) +
  geom_bar(aes (x = distribution,
                y = probability),
            stat="identity") +
geom_hline(yintercept = 0)+
theme_bw()+
labs(x = "Distribution Name",
      y = "Probability") +
facet_wrap(~outcome) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Figure 11: Probability of each outcome under different distributions



The probability of losing for Player A is generally lower across distributions, since the probability of winning each leg is 70%.

- Loses 0-2: This is the case with the lowest probability across distributions, since Player A has 70% chance of winning each leg. It is highest under AlthamBeta ($\theta = 0.50$) and BetaBinomial.
- Wins 2-0: Altham($\theta = 0.50$) and AlthamBeta($\theta = 0.50$) show the highest probability. Clustering effect makes it more likely that the player wins both first 2 legs.
- Loses 1-2 and Wins 2-1: The probability for both outcomes are higher in Altham and AlthamBeta with $\theta = 1.50$. It seems like higher theta estimates the alternating outcomes (WLL, LWL, LWW, WLW) to be more likely.
- The probability of winning and losing in the Binomial and NegBinomial are the same (0.784).

References

Wolodzko, Tymoteusz. 2026. *extraDistr: Additional Univariate and Multivariate Distributions*. https://doi.org/10.32614/CRAN.package.extraDistr.